Deploy NodeConnect (Custom Build) on DigitalOcean Kubernetes

This guide walks through deploying **NodeConnect** (a custom Activepieces build with the `ada-bmp` piece) on DigitalOcean Kubernetes (DOKS). It is based on the actual deployment to **nodesconnect.in** and includes all gotchas and fixes encountered.

Architecture

```
┌──────────────────────────────────────────────────────────────┐
│                    DigitalOcean Cloud                        │
│                                                              │
│  ┌──────────────────────────────────────────────────────┐   │
│  │             DOKS Kubernetes Cluster (nyc1)           │   │
│  │                                                      │   │
│  │   [DO Load Balancer]                                 │   │
│  │         │                                            │   │
│  │         ▼                                            │   │
│  │   [Ingress-NGINX + cert-manager (HTTPS)]            │   │
│  │         │                                            │   │
│  │         ▼                                            │   │
│  │   [NodeConnect StatefulSet - 2 replicas]             │   │
│  │     (ghcr.io/YOUR_USERNAME/nodeconnect-app:latest)   │   │
│  │         │                   │                        │   │
│  │         ▼                   ▼                        │   │
│  │   [In-cluster Redis]   [DO Managed Postgres]         │   │
│  │   (Bitnami Helm)        (external, SSL)              │   │
│  └──────────────────────────────────────────────────────┘   │
│                                                              │
│  [GHCR] ◄── Docker image pushed from local Mac              │
└──────────────────────────────────────────────────────────────┘
```

Prerequisites Summary

Tool	Purpose	Install (Mac)
`doctl`	DigitalOcean CLI	`brew install doctl`
`kubectl`	Kubernetes CLI	`brew install kubectl`
`helm`	Kubernetes package manager	`brew install helm`
`docker`	Build container images	Docker Desktop
`openssl`	Generate security keys	`brew install openssl`

Step 1: Install Required Tools

Mac (Homebrew):

```
brew install doctl kubectl helm openssl
```

Verify:

```
doctl version
kubectl version --client
helm version
docker version
```

Step 2: Create DigitalOcean Account and Connect

1. Create account at digitalocean.com and add a payment method.
2. Go to **API** in the left sidebar → **Generate New Token** → Name: `nodeconnect-deploy` → Full Access → Copy the token.
3. Connect `doctl`:

```
doctl auth init
# Paste your token when prompted
```

Verify:

```
doctl account get
```

Step 3: Build and Push Your Custom Docker Image

3a: Dockerfile Requirements

Your `Dockerfile` must include the BMP piece build step with a BuildKit cache mount. The key line is:

```
RUN --mount=type=cache,target=/root/.bun/install/cache npx nx build pieces-ada-bmp --skip-nx-cache
```

> **Why the cache mount?** The `pieces-ada-bmp` prebuild target runs `bun install`, which needs cached package manifests. Without the cache mount, `bun install` fails with `ConnectionRefused` during the Docker build because there is no network access to the bun registry for cached packages.

Also ensure the Dockerfile mirrors the nx output into the expected location:

```
RUN if [ -d dist/packages/pieces/custom/ada-bmp ]; then \
      mkdir -p packages/pieces/custom/ada-bmp/dist && \
      cp -a dist/packages/pieces/custom/ada-bmp/. packages/pieces/custom/ada-bmp/dist/; \
    fi
```

3b: Create a GitHub Personal Access Token (PAT)

1. Go to github.com/settings/tokens
2. **Generate new token (classic)** → Name: `nodeconnect-registry`
3. Scopes: `write:packages`, `read:packages`, `delete:packages`
4. Copy the token immediately.

3c: Login to GitHub Container Registry

```
# IMPORTANT: GitHub username MUST be lowercase
export GITHUB_USERNAME="your-lowercase-github-username"

echo "YOUR_GITHUB_PAT" | docker login ghcr.io -u $GITHUB_USERNAME --password-stdin
```

3d: Build the Docker Image

> **CRITICAL for Apple Silicon (M1/M2/M3 Macs):** You MUST build for `linux/amd64` because DOKS nodes run AMD64. Without `--platform linux/amd64`, the pods will crash with `exec format error`.

```
cd /path/to/your/activepieces-bmp-custom

# Step 1: Build the custom piece locally first
npx nx build pieces-ada-bmp

# Step 2: Build the Docker image for AMD64 (required even on Apple Silicon Macs)
docker build --platform linux/amd64 -t nodeconnect-app:latest -f Dockerfile .
```

This takes 15–25 minutes on first build.

3e: Tag and Push

```
docker tag nodeconnect-app:latest ghcr.io/$GITHUB_USERNAME/nodeconnect-app:latest
docker push ghcr.io/$GITHUB_USERNAME/nodeconnect-app:latest
```

3f: Verify on GitHub

Go to `https://github.com/users/YOUR_USERNAME/packages/container/package/nodeconnect-app` and confirm the image is listed. Keep it **Private** (recommended).

Step 4: Create Your Kubernetes Cluster

```
doctl kubernetes cluster create nodeconnect-cluster \
  --region nyc1 \
  --node-pool "name=worker-pool;size=s-2vcpu-4gb;count=2" \
  --version latest
```

This takes 5–10 minutes. Once done, connect `kubectl`:

```
doctl kubernetes cluster kubeconfig save nodeconnect-cluster
```

Verify:

```
kubectl get nodes
# Should show 2 nodes with STATUS "Ready"
```

Step 5: Create Databases

5a: Create Managed PostgreSQL

```
doctl databases create nodeconnect-postgres \
  --engine pg \
  --region nyc1 \
  --size db-s-1vcpu-1gb \
  --num-nodes 1 \
  --version 16
```

Wait 3–5 minutes, then get connection details:

```
doctl databases connection nodeconnect-postgres
```

Save: `Host` , `Port` , `User` , `Password` , `Database` .

> **SSL Note:** DigitalOcean managed Postgres requires SSL. We handle this with `AP_POSTGRES_USE_SSL=true` plus `NODE_TLS_REJECT_UNAUTHORIZED=0` (see Step 7).
> For production, download the CA certificate from the DO dashboard and set `AP_POSTGRES_SSL_CA` .

5b: Deploy In-Cluster Redis (Bitnami Helm Chart)

> **Why in-cluster?** DigitalOcean managed Redis is not available in all regions (e.g., `nyc1`).
> An in-cluster Redis via Bitnami is simple and avoids region availability issues.

```
# Create the namespace first (shared with the app)
kubectl create namespace nodeconnect

# Add Bitnami repo
helm repo add bitnami https://charts.bitnami.com/bitnami
helm repo update

# Install Redis (use SINGLE quotes around password to avoid bash expansion of special chars)
helm install nodeconnect-redis bitnami/redis \
  --namespace nodeconnect \
  --set auth.password='YourSecureRedisPassword!' \
  --set architecture=standalone \
  --set master.persistence.size=2Gi
```

Redis connection details for secrets:

Setting	Value
`AP_REDIS_HOST`	`nodeconnect-redis-master.nodeconnect.svc.cluster.local`
`AP_REDIS_PORT`	`6379`
`AP_REDIS_PASSWORD`	(the password you set above)
`AP_REDIS_USE_SSL`	`false`
`AP_REDIS_USER`	`default`
`AP_REDIS_TYPE`	`DEFAULT`

Step 6: Set Up Ingress Controller and HTTPS

6a: Install Ingress-NGINX

```
helm repo add ingress-nginx https://kubernetes.github.io/ingress-nginx
helm repo update

helm install ingress-nginx ingress-nginx/ingress-nginx \
  --namespace ingress-nginx \
  --create-namespace \
  --set controller.service.annotations."service\.beta\.kubernetes\.io/do-loadbalancer-name"="nodeconnect-lb" \
  --set controller.service.annotations."service\.beta\.kubernetes\.io/do-loadbalancer-protocol"="http"
```

6b: Wait for Load Balancer IP

```
kubectl get svc -n ingress-nginx ingress-nginx-controller --watch
# Wait until EXTERNAL-IP changes from <pending> to an actual IP. Press Ctrl+C to stop.
```

Save this IP — you need it for DNS (Step 9).

6c: Install cert-manager (for HTTPS/Let's Encrypt)

```
kubectl apply -f https://github.com/cert-manager/cert-manager/releases/download/v1.13.0/cert-manager.yaml

# Wait for it to be ready
kubectl wait --for=condition=Available deployment --all -n cert-manager --timeout=300s
```

6d: Create the ClusterIssuer

```
cat << 'EOF' | kubectl apply -f -
apiVersion: cert-manager.io/v1
kind: ClusterIssuer
metadata:
  name: letsencrypt-prod
spec:
  acme:
    server: https://acme-v02.api.letsencrypt.org/directory
    email: your-email@example.com
    privateKeySecretRef:
      name: letsencrypt-prod
    solvers:
    - http01:
        ingress:
          class: nginx
EOF
```

Step 7: Create Kubernetes Secrets

7a: Generate Security Keys

```
echo "AP_ENCRYPTION_KEY: $(openssl rand -hex 16)"
echo "AP_JWT_SECRET: $(openssl rand -hex 32)"
echo "AP_WORKER_TOKEN: $(openssl rand -hex 32)"
```

Save all three values.

7b: Set Environment Variables

```
# PostgreSQL (from Step 5a)
export DB_HOST="your-db-xxxxx.db.ondigitalocean.com"
export DB_PORT="25060"
export DB_USER="doadmin"
export DB_PASSWORD="AVNS_xxxxxxxxxxxxxxx"
export DB_NAME="defaultdb"

# Redis (from Step 5b)
export REDIS_HOST="nodeconnect-redis-master.nodeconnect.svc.cluster.local"
export REDIS_PORT="6379"
export REDIS_PASSWORD="YourSecureRedisPassword!"

# Security keys (from 7a)
export ENCRYPTION_KEY="paste-here"
export JWT_SECRET="paste-here"
export WORKER_TOKEN="paste-here"

# Domain
export MY_DOMAIN="nodesconnect.in"

# GitHub (lowercase)
export GITHUB_USERNAME="your-lowercase-github-username"
```

7c: Create the Image Pull Secret

```
kubectl create secret docker-registry ghcr-secret \
  --namespace nodeconnect \
  --docker-server=ghcr.io \
  --docker-username=$GITHUB_USERNAME \
  --docker-password=YOUR_GITHUB_PAT \
  --docker-email=your-email@example.com
```

7d: Create All Application Secrets

```
# Config secrets (includes BMP and custom piece config)
kubectl create secret generic nodeconnect-config-secrets \
  --namespace nodeconnect \
  --from-literal=AP_EDITION=ce \
  --from-literal=AP_EXECUTION_MODE=UNSANDBOXED \
  --from-literal=AP_ENVIRONMENT=production \
  --from-literal=AP_PORT=3001 \
  --from-literal=AP_FRONTEND_URL=https://$MY_DOMAIN \
  --from-literal=AP_ENABLE_FLOW_ON_PUBLISH=true \
  --from-literal=AP_CLIENT_REAL_IP_HEADER=X-Forwarded-For \
  --from-literal=AP_DB_TYPE=POSTGRES \
  --from-literal=AP_ENGINE_EXECUTABLE_PATH="dist/packages/engine/main.js" \
  --from-literal=AP_PIECES_SYNC_MODE=OFFICIAL_AUTO \
  --from-literal=AP_PIECES_SOURCE=FILE \
  --from-literal=AP_DEV_PIECES=ada-bmp \
  --from-literal=AP_BMP_ENABLED=true \
  --from-literal=AP_BMP_ORGANIZATIONS=true \
  --from-literal=AP_BMP_SUPER_ADMIN=true \
  --from-literal=AP_BMP_ACCOUNT_SWITCHING=true \
  --from-literal=AP_MULTI_TENANT_MODE=true \
  --from-literal=ADA_BMP_API_URL=https://bmpapi.bmp.ada-asia.my \
  --from-literal=ADA_BMP_TIMEOUT=30000 \
  --from-literal=AP_SANDBOX_PROPAGATED_ENV_VARS=ADA_BMP_API_URL \
  --from-literal=AP_WEBHOOK_TIMEOUT_SECONDS=30 \
  --from-literal=AP_TRIGGER_DEFAULT_POLL_INTERVAL=1 \
  --from-literal=AP_PLATFORM_ID_FOR_DEDICATED_WORKER=""

# Auth secrets
kubectl create secret generic nodeconnect-auth-secrets \
  --namespace nodeconnect \
  --from-literal=AP_ENCRYPTION_KEY=$ENCRYPTION_KEY \
  --from-literal=AP_JWT_SECRET=$JWT_SECRET \
  --from-literal=AP_API_KEY="" \
  --from-literal=AP_WORKER_TOKEN=$WORKER_TOKEN \
  --from-literal=AP_GOOGLE_CLIENT_ID="" \
  --from-literal=AP_GOOGLE_CLIENT_SECRET="" \
  --from-literal=AP_FIREBASE_HASH_PARAMETERS=""

# Database secrets
kubectl create secret generic nodeconnect-db-secrets \
  --namespace nodeconnect \
  --from-literal=AP_POSTGRES_DATABASE=$DB_NAME \
  --from-literal=AP_POSTGRES_HOST=$DB_HOST \
  --from-literal=AP_POSTGRES_PORT=$DB_PORT \
  --from-literal=AP_POSTGRES_PASSWORD=$DB_PASSWORD \
  --from-literal=AP_POSTGRES_USERNAME=$DB_USER \
  --from-literal=AP_POSTGRES_USE_SSL=true \
  --from-literal=AP_POSTGRES_POOL_SIZE="" \
  --from-literal=AP_POSTGRES_SSL_CA=""

# Redis secrets (in-cluster, no SSL)
kubectl create secret generic nodeconnect-redis-secrets \
  --namespace nodeconnect \
  --from-literal=AP_REDIS_TYPE=DEFAULT \
  --from-literal=AP_REDIS_HOST=$REDIS_HOST \
  --from-literal=AP_REDIS_PORT=$REDIS_PORT \
  --from-literal=AP_REDIS_PASSWORD=$REDIS_PASSWORD \
  --from-literal=AP_REDIS_USE_SSL=false \
  --from-literal=AP_REDIS_USER=default
```

```
# Queue secrets
kubectl create secret generic nodeconnect-queue-secrets \
  --namespace nodeconnect \
  --from-literal=AP_QUEUE_MODE=REDIS \
  --from-literal=AP_QUEUE_UI_ENABLED=false \
  --from-literal=AP_QUEUE_UI_USERNAME="" \
  --from-literal=AP_QUEUE_UI_PASSWORD=""

# Log secrets
kubectl create secret generic nodeconnect-log-secrets \
  --namespace nodeconnect \
  --from-literal=AP_LOG_LEVEL=info \
  --from-literal=AP_LOG_PRETTY=false \
  --from-literal=AP_SENTRY_DSN="" \
  --from-literal=AP_HYPERDX_TOKEN=""

# Telemetry secrets
# IMPORTANT: AP_OTEL_ENABLED must be explicitly "false", NOT an empty string
kubectl create secret generic nodeconnect-telemetry-secrets \
  --namespace nodeconnect \
  --from-literal=AP_TELEMETRY_ENABLED=false \
  --from-literal=AP_OTEL_ENABLED=false \
  --from-literal=AP_FEATUREBASE_API_KEY=""

# Remaining secrets (required by the Helm chart, can be empty)
for secret in nodeconnect-limits-secrets nodeconnect-cloudflare-secrets \
  nodeconnect-integrations-secrets nodeconnect-s3-secrets \
  nodeconnect-smtp-secrets nodeconnect-stripe-secrets nodeconnect-otel-secrets; do
  kubectl create secret generic $secret --namespace nodeconnect --from-literal=placeholder=""
done
```

> **Gotcha — AP_OTEL_ENABLED:** This variable expects exactly `"true"` or `"false"`. An empty string causes the app to crash with `EnvVarError`. Always set it explicitly to `false`.

Verify:

```
kubectl get secrets -n nodeconnect
```

Step 8: Deploy with Helm

8a: Navigate to Your Helm Chart

Use your local repository that already contains the Activepieces Helm chart:

```
cd /path/to/your/activepieces-bmp-custom/deploy/activepieces-helm
```

8b: Create `values-digitalocean.yaml`

Create this file in the Helm chart directory. Replace placeholders with your actual values:

```yaml
replicaCount: 2

image:
  repository: ghcr.io/YOUR_GITHUB_USERNAME/nodeconnect-app
  pullPolicy: Always
  tag: "latest"

imagePullSecrets:
  - name: ghcr-secret

workloadType: statefulset

container:
  port: 3001

resources:
  limits:
    cpu: "1000m"
    memory: "2Gi"
  requests:
    cpu: "250m"
    memory: "512Mi"

persistence:
  enabled: true
  size: 10Gi
  storageClassName: do-block-storage
  mountPath: "/usr/src/app/cache"

service:
  type: ClusterIP
  port: 80

ingress:
  enabled: true
  className: nginx
  annotations:
    cert-manager.io/cluster-issuer: letsencrypt-prod
    nginx.ingress.kubernetes.io/proxy-body-size: "100m"
    nginx.ingress.kubernetes.io/proxy-read-timeout: "900"
    nginx.ingress.kubernetes.io/proxy-send-timeout: "900"
    nginx.ingress.kubernetes.io/proxy-connect-timeout: "60"
  hosts:
    - host: YOUR_DOMAIN
      paths:
        - path: /
          pathType: Prefix
  tls:
    - secretName: nodeconnect-tls
      hosts:
        - YOUR_DOMAIN

activepieces:
  frontendUrl: "https://YOUR_DOMAIN"
  edition: "ce"
  executionMode: "UNSANDBOXED"
  environment: "prod"
  telemetryEnabled: false

postgresql:
  enabled: false
```

```yaml
redis:
  enabled: false

autoscaling:
  enabled: false

# NODE_TLS_REJECT_UNAUTHORIZED=0 bypasses self-signed cert errors for DO managed Postgres.
# For production, download the CA cert from DO dashboard and use AP_POSTGRES_SSL_CA instead.
activepiecesConfig:
  AP_CONTAINER_TYPE: ""
  NODE_TLS_REJECT_UNAUTHORIZED: "0"

# IMPORTANT: Null out ALL default activepieces-*-secrets from values.yaml.
# Without this, Helm merges both sets and Kubernetes rejects the deployment
# with "duplicate entries for key" errors.
activepiecesEnvVariables:
  activepieces-config-secrets: null
  activepieces-auth-secrets: null
  activepieces-queue-secrets: null
  activepieces-log-secrets: null
  activepieces-telemetry-secrets: null
  activepieces-limits-secrets: null
  activepieces-cloudflare-secrets: null
  activepieces-integrations-secrets: null
  activepieces-db-secrets: null
  activepieces-redis-secrets: null
  activepieces-s3-secrets: null
  activepieces-smtp-secrets: null
  activepieces-stripe-secrets: null
  activepieces-otel-secrets: null

  nodeconnect-config-secrets:
    - AP_EDITION
    - AP_EXECUTION_MODE
    - AP_ENVIRONMENT
    - AP_PORT
    - AP_FRONTEND_URL
    - AP_ENABLE_FLOW_ON_PUBLISH
    - AP_CLIENT_REAL_IP_HEADER
    - AP_DB_TYPE
    - AP_ENGINE_EXECUTABLE_PATH
    - AP_PLATFORM_ID_FOR_DEDICATED_WORKER
    - AP_PIECES_SYNC_MODE
    - AP_PIECES_SOURCE
    - AP_DEV_PIECES
    - AP_BMP_ENABLED
    - AP_BMP_ORGANIZATIONS
    - AP_BMP_SUPER_ADMIN
    - AP_BMP_ACCOUNT_SWITCHING
    - AP_MULTI_TENANT_MODE
    - ADA_BMP_API_URL
    - ADA_BMP_TIMEOUT
    - AP_SANDBOX_PROPAGATED_ENV_VARS
    - AP_WEBHOOK_TIMEOUT_SECONDS
    - AP_TRIGGER_DEFAULT_POLL_INTERVAL
  nodeconnect-auth-secrets:
    - AP_ENCRYPTION_KEY
    - AP_JWT_SECRET
    - AP_API_KEY
    - AP_WORKER_TOKEN
    - AP_GOOGLE_CLIENT_ID
```

```yaml
  - AP_GOOGLE_CLIENT_SECRET
  - AP_FIREBASE_HASH_PARAMETERS
nodeconnect-queue-secrets:
  - AP_QUEUE_MODE
  - AP_QUEUE_UI_ENABLED
  - AP_QUEUE_UI_USERNAME
  - AP_QUEUE_UI_PASSWORD
nodeconnect-log-secrets:
  - AP_LOG_LEVEL
  - AP_LOG_PRETTY
  - AP_SENTRY_DSN
  - AP_HYPERDX_TOKEN
nodeconnect-telemetry-secrets:
  - AP_TELEMETRY_ENABLED
  - AP_OTEL_ENABLED
  - AP_FEATUREBASE_API_KEY
nodeconnect-limits-secrets:
  - AP_MAX_CONCURRENT_JOBS_PER_PROJECT
  - AP_PROJECT_RATE_LIMITER_ENABLED
  - AP_API_RATE_LIMIT_AUTHN_ENABLED
  - AP_MAX_FILE_SIZE_MB
  - AP_MAX_FLOW_RUN_LOG_SIZE_MB
  - AP_EXECUTION_DATA_RETENTION_DAYS
nodeconnect-cloudflare-secrets:
  - AP_CLOUDFLARE_API_BASE
  - AP_CLOUDFLARE_ZONE_ID
  - AP_CLOUDFLARE_API_TOKEN
nodeconnect-integrations-secrets:
  - AP_APP_WEBHOOK_SECRETS
  - AP_SECRET_MANAGER_API_KEY
  - AP_OPENROUTER_PROVISION_KEY
  - AP_TEMPLATES_API_KEY
  - AP_TEMPLATE_MANAGER_API_KEY
nodeconnect-db-secrets:
  - AP_POSTGRES_DATABASE
  - AP_POSTGRES_HOST
  - AP_POSTGRES_PORT
  - AP_POSTGRES_PASSWORD
  - AP_POSTGRES_USERNAME
  - AP_POSTGRES_USE_SSL
  - AP_POSTGRES_POOL_SIZE
  - AP_POSTGRES_SSL_CA
nodeconnect-redis-secrets:
  - AP_REDIS_TYPE
  - AP_REDIS_HOST
  - AP_REDIS_PORT
  - AP_REDIS_PASSWORD
  - AP_REDIS_USE_SSL
  - AP_REDIS_USER
nodeconnect-s3-secrets:
  - AP_FILE_STORAGE_LOCATION
  - AP_S3_BUCKET
  - AP_S3_REGION
  - AP_S3_ENDPOINT
  - AP_S3_USE_SIGNED_URLS
  - AP_S3_ACCESS_KEY_ID
  - AP_S3_SECRET_ACCESS_KEY
nodeconnect-smtp-secrets:
  - AP_SMTP_HOST
  - AP_SMTP_PORT
  - AP_SMTP_USERNAME
  - AP_SMTP_PASSWORD
```

```
    - AP_SMTP_SENDER_EMAIL
    - AP_SMTP_SENDER_NAME
  nodeconnect-stripe-secrets:
    - AP_STRIPE_SECRET_KEY
    - AP_STRIPE_WEBHOOK_SECRET
  nodeconnect-otel-secrets:
    - OTEL_EXPORTER_OTLP_ENDPOINT
    - OTEL_EXPORTER_OTLP_HEADERS
```

8c: Update Helm Dependencies and Deploy

```
helm dependency update

helm install nodeconnect . \
  --namespace nodeconnect \
  --values values-digitalocean.yaml
```

Expected output:

```
NAME: nodeconnect
LAST DEPLOYED: ...
NAMESPACE: nodeconnect
STATUS: deployed
```

8d: Watch Pods Start

```
kubectl get pods -n nodeconnect --watch
```

Wait until all pods show `1/1 Running`. Press Ctrl+C to stop watching.

> **If pods show `CrashLoopBackOff`:**

```
> kubectl logs -n nodeconnect nodeconnect-activepieces-0
>
```

> Common causes:
>
> - `exec format error` → rebuild image with `--platform linux/amd64`
> - `EnvVarError: "AP_OTEL_ENABLED"` → recreate `nodeconnect-telemetry-secrets` with `AP_OTEL_ENABLED=false`
> - `self-signed certificate in certificate chain` → ensure `NODE_TLS_REJECT_UNAUTHORIZED: "0"` is in `activepiecesConfig`

Step 9: Configure Your Domain (DNS)

9a: Get Load Balancer IP

```
kubectl get svc -n ingress-nginx ingress-nginx-controller \
  -o jsonpath='{.status.loadBalancer.ingress[0].ip}'
```

9b: Configure DNS Records

At your DNS provider (e.g., Cloudflare), create:

Type	Name	Value	Proxy
A	@	LOAD_BALANCER_IP	Off
A	www	LOAD_BALANCER_IP	Off

> **Cloudflare users:** Turn proxy **off** (DNS only / grey cloud) so cert-manager can complete the HTTP-01 challenge. You can enable proxy after the certificate is issued.

9c: Wait for DNS Propagation

```
nslookup YOUR_DOMAIN
# Should resolve to your load balancer IP
```

DNS propagation typically takes 5–30 minutes.

Step 10: Verify Everything is Working

10a: Check Pods

```
kubectl get pods -n nodeconnect
# All pods should be 1/1 Running
```

10b: Check Live Logs

```
kubectl logs -n nodeconnect nodeconnect-activepieces-0 -f
```

Look for:

```
info -- [ServerApi#start] 🚀 App is listening at 0.0.0.0:3001
```

10c: Check the Ingress and Certificate

```
kubectl get ingress -n nodeconnect
kubectl get certificate -n nodeconnect
# READY should be True
```

10d: Verify Custom Piece is Loaded

```
# Check piece files exist in the container
kubectl exec -n nodeconnect nodeconnect-activepieces-0 -- \
  ls -la /usr/src/app/packages/pieces/custom/ada-bmp/dist/

# Check env vars are correct
kubectl exec -n nodeconnect nodeconnect-activepieces-0 -- \
  env | grep -E "AP_PIECES_SOURCE|AP_DEV_PIECES"
```

Expected:

```
AP_PIECES_SOURCE=FILE
AP_DEV_PIECES=ada-bmp
```

10e: Access the Application

Open `https://YOUR_DOMAIN` in your browser. Create your first admin account, then create a flow and verify the **ada-bmp** piece appears in the piece selector.

How to Update After Code Changes

Rebuild and Push

```
cd /path/to/your/activepieces-bmp-custom

# Build the custom piece first
npx nx build pieces-ada-bmp

# Build the Docker image for AMD64 (always required for DOKS)
docker build --platform linux/amd64 -t nodeconnect-app:latest -f Dockerfile .

# Tag and push
docker tag nodeconnect-app:latest ghcr.io/$GITHUB_USERNAME/nodeconnect-app:latest
docker push ghcr.io/$GITHUB_USERNAME/nodeconnect-app:latest
```

Deploy the Update

Option A — Restart pods (if using `latest` tag):

```
kubectl rollout restart statefulset nodeconnect-activepieces -n nodeconnect
kubectl rollout status statefulset nodeconnect-activepieces -n nodeconnect
```

Option B — Helm upgrade (if you changed values):

```
cd /path/to/your/activepieces-bmp-custom/deploy/activepieces-helm

helm upgrade nodeconnect . \
  --namespace nodeconnect \
  --values values-digitalocean.yaml
```

Update a Kubernetes Secret

To change a secret (e.g., `nodeconnect-config-secrets`):

```
# Delete the old one
kubectl delete secret nodeconnect-config-secrets -n nodeconnect

# Recreate with updated values
kubectl create secret generic nodeconnect-config-secrets \
  --namespace nodeconnect \
  --from-literal=KEY1=value1 \
  --from-literal=KEY2=value2 \
  ...

# Restart pods to pick up changes
kubectl delete pod nodeconnect-activepieces-0 nodeconnect-activepieces-1 -n nodeconnect
```

Troubleshooting Reference

Symptom	Cause	Fix
`exec format error` in pod logs	Image built for ARM64, DOKS needs AMD64	Rebuild with `--platform linux/amd64`
`EnvVarError: "AP_OTEL_ENABLED"`	Empty string instead of `false`	Recreate `nodeconnect-telemetry-secrets` with `AP_OTEL_ENABLED=false`
`self-signed certificate in certificate chain`	DO managed Postgres uses self-signed SSL	Set `NODE_TLS_REJECT_UNAUTHORIZED: "0"` in `activepiecesConfig`
`duplicate entries for key` on helm install	Default secrets from `values.yaml` merge with custom secrets	Null out all `activepieces-*-secrets` in your values override
`cannot reuse a name that is still in use`	Previous failed Helm release still exists	`helm uninstall nodeconnect -n nodeconnect` then re-install
`CrashLoopBackOff`	Various — check logs	`kubectl logs -n nodeconnect nodeconnect-activepieces-0`
Custom piece not visible	`AP_PIECES_SOURCE` / `AP_DEV_PIECES` not set	Set `AP_PIECES_SOURCE=FILE` and `AP_DEV_PIECES=ada-bmp`
Redis password with `!` causes `dquote>`	Bash history expansion	Use single quotes: `--set auth.password='Pass!'`
`ConnectionRefused` during Docker build (bun install)	Missing BuildKit cache mount	Add `--mount=type=cache,target=/root/.bun/install/cache` to the RUN line

Useful Commands

```
# View all resources in the namespace
kubectl get all -n nodeconnect

# Live logs for a pod
kubectl logs -n nodeconnect nodeconnect-activepieces-0 -f

# Exec into a pod for debugging
kubectl exec -it -n nodeconnect nodeconnect-activepieces-0 -- /bin/sh

# Check all env vars in a pod
kubectl exec -n nodeconnect nodeconnect-activepieces-0 -- env | sort

# Restart all app pods
kubectl rollout restart statefulset nodeconnect-activepieces -n nodeconnect

# Delete and reinstall via Helm
helm uninstall nodeconnect -n nodeconnect
helm install nodeconnect . --namespace nodeconnect --values values-digitalocean.yaml

# Port-forward for local testing (access at http://localhost:8080)
kubectl port-forward -n nodeconnect svc/nodeconnect-activepieces 8080:80
```

Monthly Costs (DigitalOcean)

Resource	Cost
DOKS Nodes (2x 2vCPU/4GB)	$48
Managed PostgreSQL	$15
In-cluster Redis	$0 (runs on existing nodes)
Load Balancer	$12
Block Storage (20GB)	$2
Total	**~$77/mo**

Next Steps

1. **Database backups** — Enable automatic backups for PostgreSQL in the DO dashboard.
2. **Proper SSL for Postgres** — Download the DO CA certificate and set `AP_POSTGRES_SSL_CA` instead of `NODE_TLS_REJECT_UNAUTHORIZED=0`.
3. **Monitoring** — Add DigitalOcean Monitoring or Prometheus/Grafana.
4. **Scaling** — Enable `autoscaling` in `values-digitalocean.yaml` for auto-scaling based on CPU.
5. **CI/CD** — Set up GitHub Actions to auto-build and deploy on push to main.